DASC 3240, L11

More plots – Time series, dendrogram, network, and map

2023/24 SPRING SEMESTER

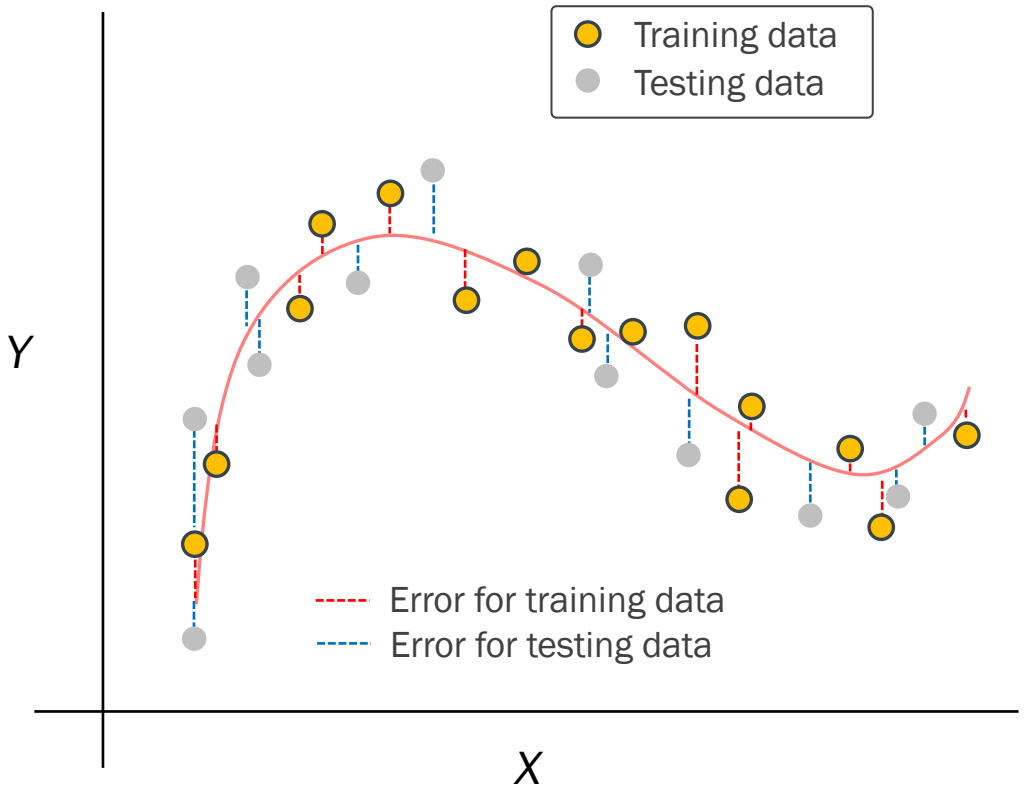
MASAYUKI USHIO, ASSISTANT PROFESSOR AT OCES
USHIO@UST.HK (RM CYT-2013)

Feedback on the previous assignments

- LOWESS fitting curve and enumerate function
  - Please check “L10_ScatterPlot-Example.jpynb” in the GitHub repository.
- Which `frac` is the most appropriate?
  - Keywords are: cross-validation, errors, training data and testing data
  - Machine Learning and Cross-Validation
    - https://youtu.be/fSytzGwwBVw?si=rA47Jpj4Y4WRK1vG
  - K-Fold Cross-Validation
    - https://youtu.be/kituDjzXwfE?si=Te060Pi7Qg4S1WSM

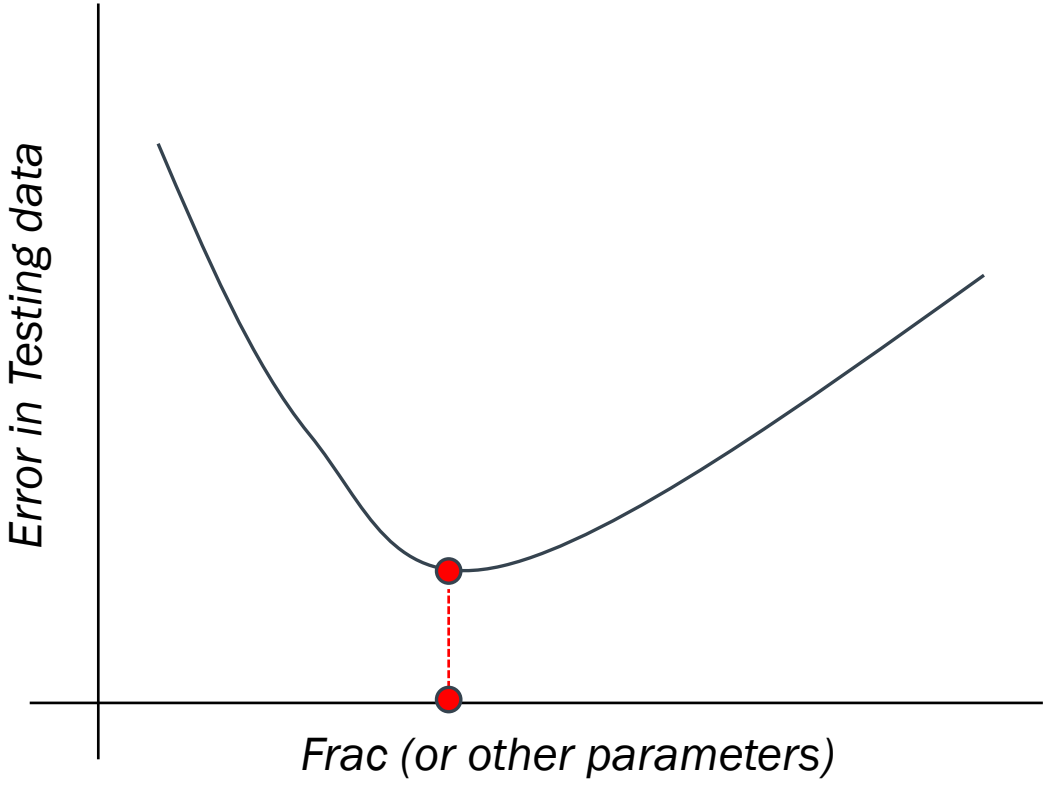
Feedback on the previous assignments

- Cross-validation
  - Divide the data into two categories: Training data and Testing data. You construct a model (or regression) using **the training data**, and evaluate its performance (errors) using **the testing data**.



Feedback on the previous assignments

- Cross-validation
  - Divide the data into two categories: Training data and Testing data. You construct a model (or regression) using **the training data**, and evaluate its performance (errors) using **the testing data**.



Today’s outline

- More plots
  1. Time series
  2. Dendrogram
  3. Network
  4. Map

More plots

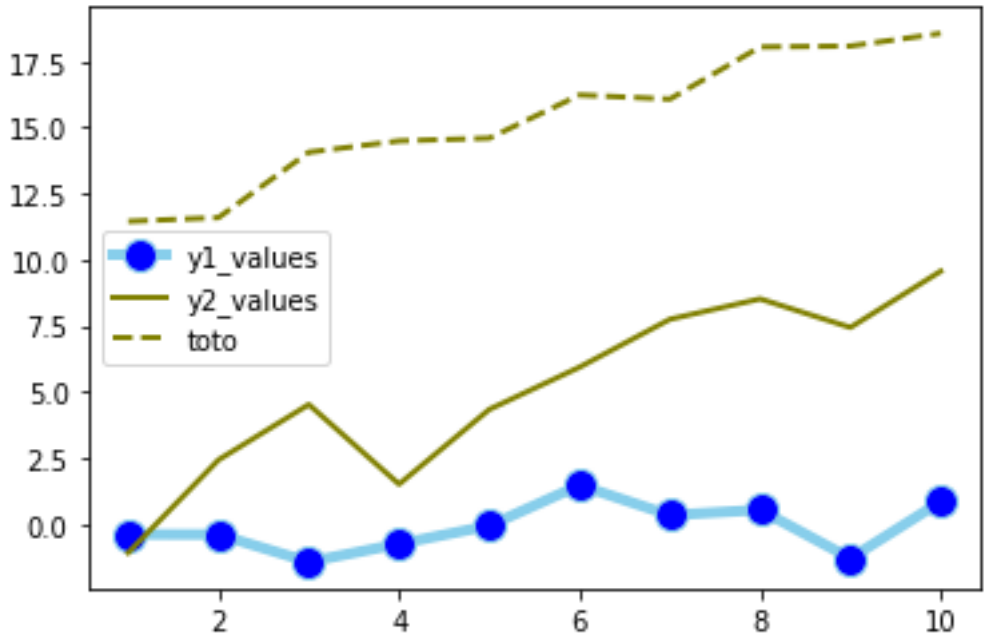
TIME SERIES, DENDROGRAM, NETWORK, AND MAP

Time series

Time series

- **Time series plots** (or time series, time series charts, etc.) refer to all charts representing the evolution of a numeric value (normally) along time.
- Line chart, scatter plot, barplot, boxplot, and area chart can be used for time series visualization.
- Line chart or line chart with symbol is one of the most common methods for time series visualization.

Linechart for time series data

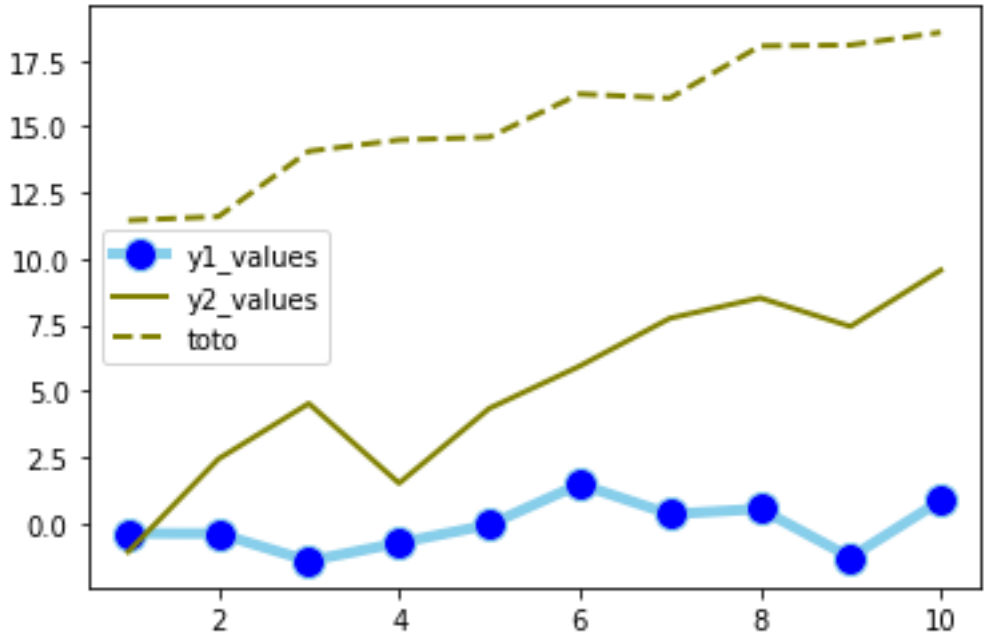


Time (e.g., day or hour)

Time series: Some considerations

- "**Date**" is a bit tricky variable. We need some caution to handle "date" variables.
- "**Log-transformation**" is often used for visualization purpose not only in time series, but also in other plots. It is useful but sometimes need some caution.
- **Interpretation of the trend**: we need to be careful.

Linechart for time series data



Time (e.g., day or hour)

“Date” variable

- Date and time are described using characters: e.g., “2024-03-01”.
- We have to let the computer know that “2024-03-01” is **NOT** characters, but “date”.
- **This is critically important** especially when we do statistical analysis using date/time variables. Otherwise, the computer regard the date strings as “character” (= categorical variable), and the results will be wrong.
- Please always check if the date variables are handled as “date”, not “character”.

“Date” variable

- Another issue is “date” and “time” have **many ways** to be described: e.g., “2024-03-01”, “2024/03/01”, “1 Mar 2024”, …
- When we want to plot (and analyze) time series data, we should always use **a consistent way** to describe date/time.
- This is a daunting task; **we need a package!**
- (By the way, Microsoft Excel often changes “date” format automatically… You should be careful if you want to edit your CSV file using Microsoft Excel!!)

datetime package

datetime — Basic date and time types

Source code: Lib/datetime.py

- The “datetime” module is a built-in module in Python.

If you simply type “2024-01-17 11:33:34.293482”, it must be “str”. But ”now” is “datetime.datetime”.

For more detail, see the hands-on items.

```python
# libraries
import datetime

# create a datetime object for the current date and time
now = datetime.datetime.now()

print(now)
```

```
2024-01-17 11:33:34.293482
```

```python
# Check the type of now
type(now)
```

```
datetime.datetime
```

pandas package

- The “pandas” package also has a similar function.

> API reference > General functions > pandas.to_datetime

pandas.to_datetime

```
pandas.to_datetime(arg, errors='raise', dayfirst=False, yearfirst=False,
utc=False, format=None, exact=_NoDefault.no_default, unit=None,
infer_datetime_format=_NoDefault.no_default, origin='unix', cache=True)
```

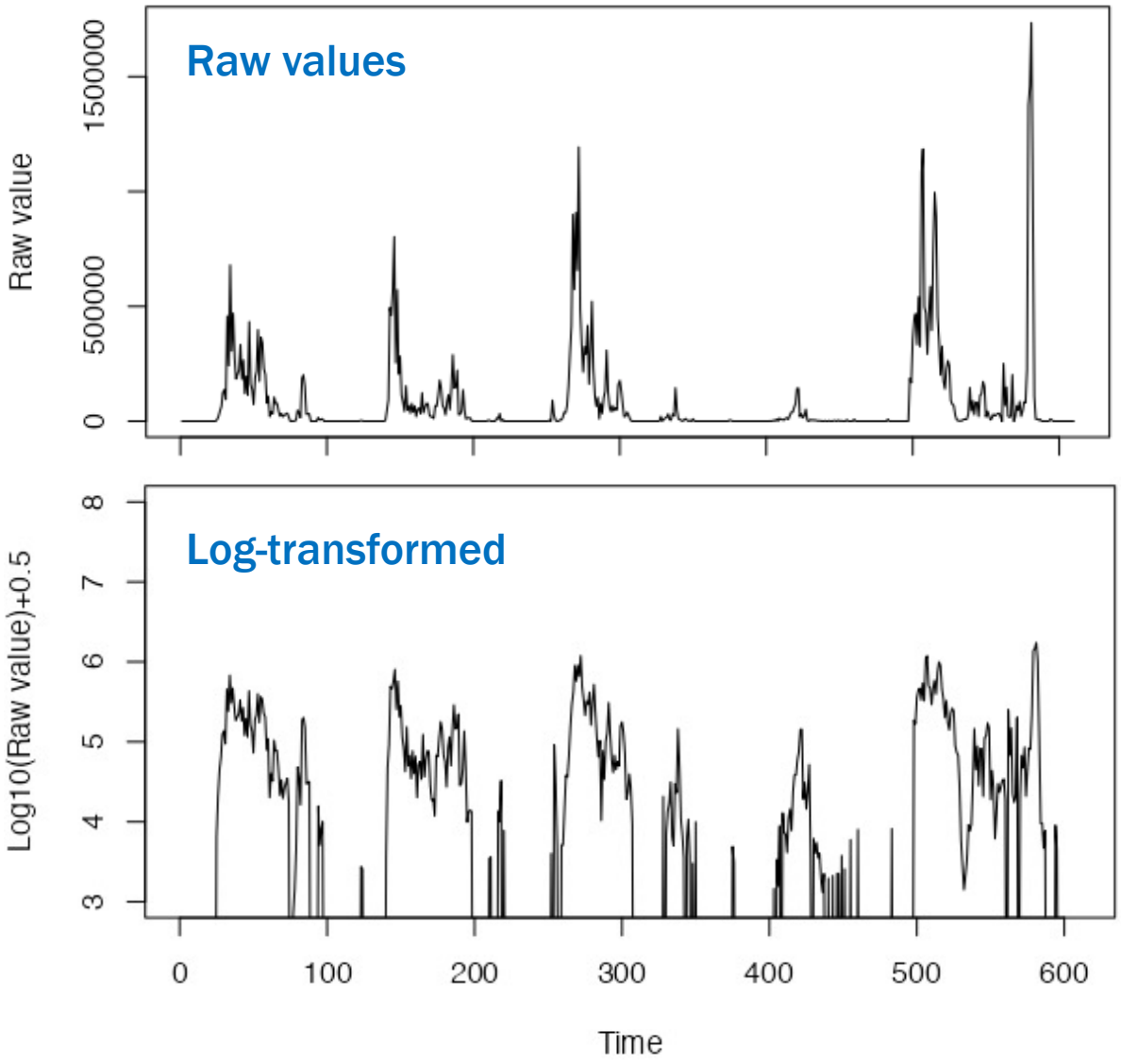
Convert argument to datetime. [source]

This function converts a scalar, array-like, `Series` or `DataFrame`/dict-like to a pandas datetime object.

https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.to_datetime.html

Log-transformation

- In time series data, we sometimes observe an exponential increase.
- Under such circumstances, "**log-transformation**" is useful to effectively visualize the pattern.
- In statistical analysis, log-transformation is commonly used to ensure "the normality" of the data (not explained in this course).



Log-transformation

- However, if the values include **0 or negative values**, we cannot simply log-transform them because we cannot define **log(0)**.
- A positive value (e.g., +1 or +0.5) may be added to enable log-transformation. (But, 1, 0.5, or other values? See https://doi.org/10.1007/s101440050026)
- Also, please clarify which base (2, e, 10, or other?) you used.

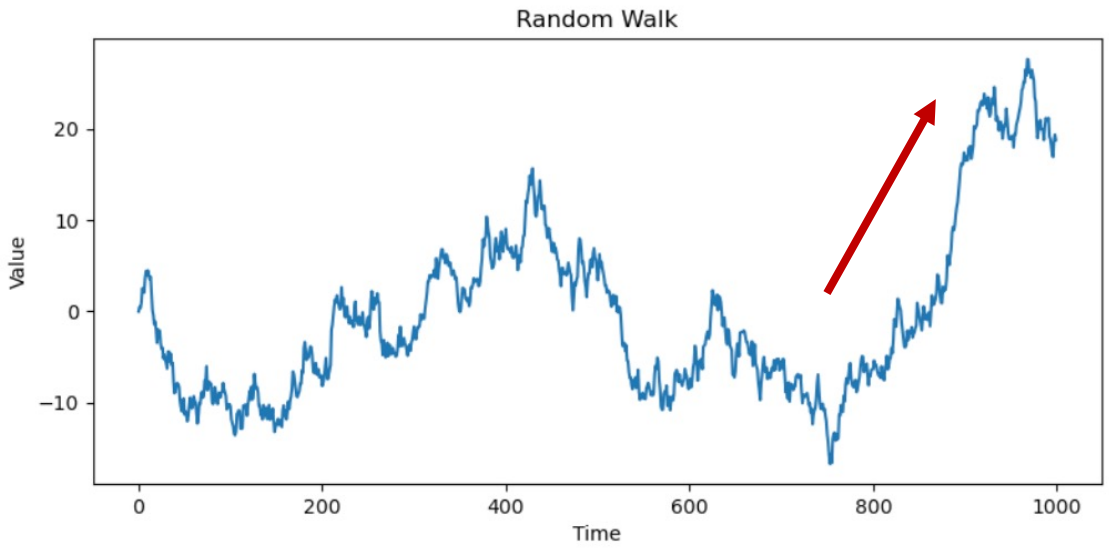
Log-transformation

StatQuest (https://statquest.org/)

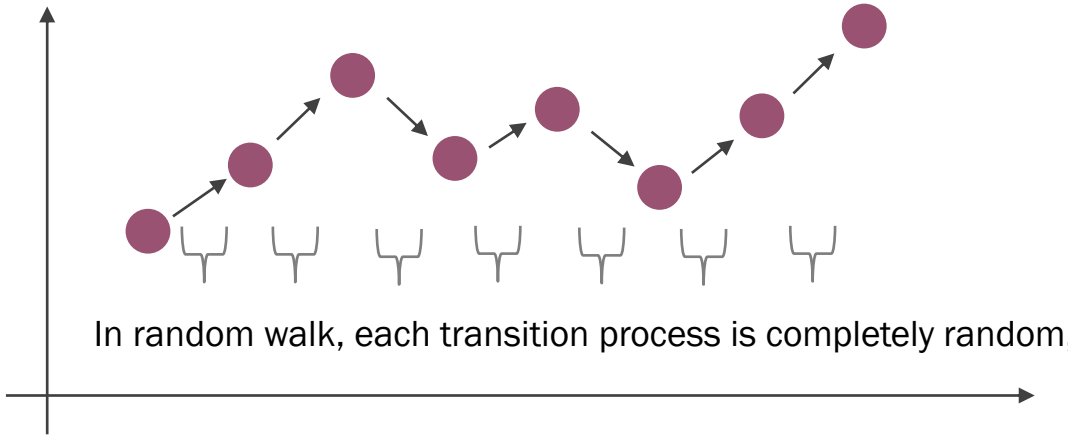
https://youtu.be/VSi0Z04fWj0?si=CBgaGOVxJeuf5CzX

Interpretation of the trend

- If you find a trend in time series, do you think there is a specific mechanism to generate it?
- Not always. Random walk can generate the trend without a specific mechanism.



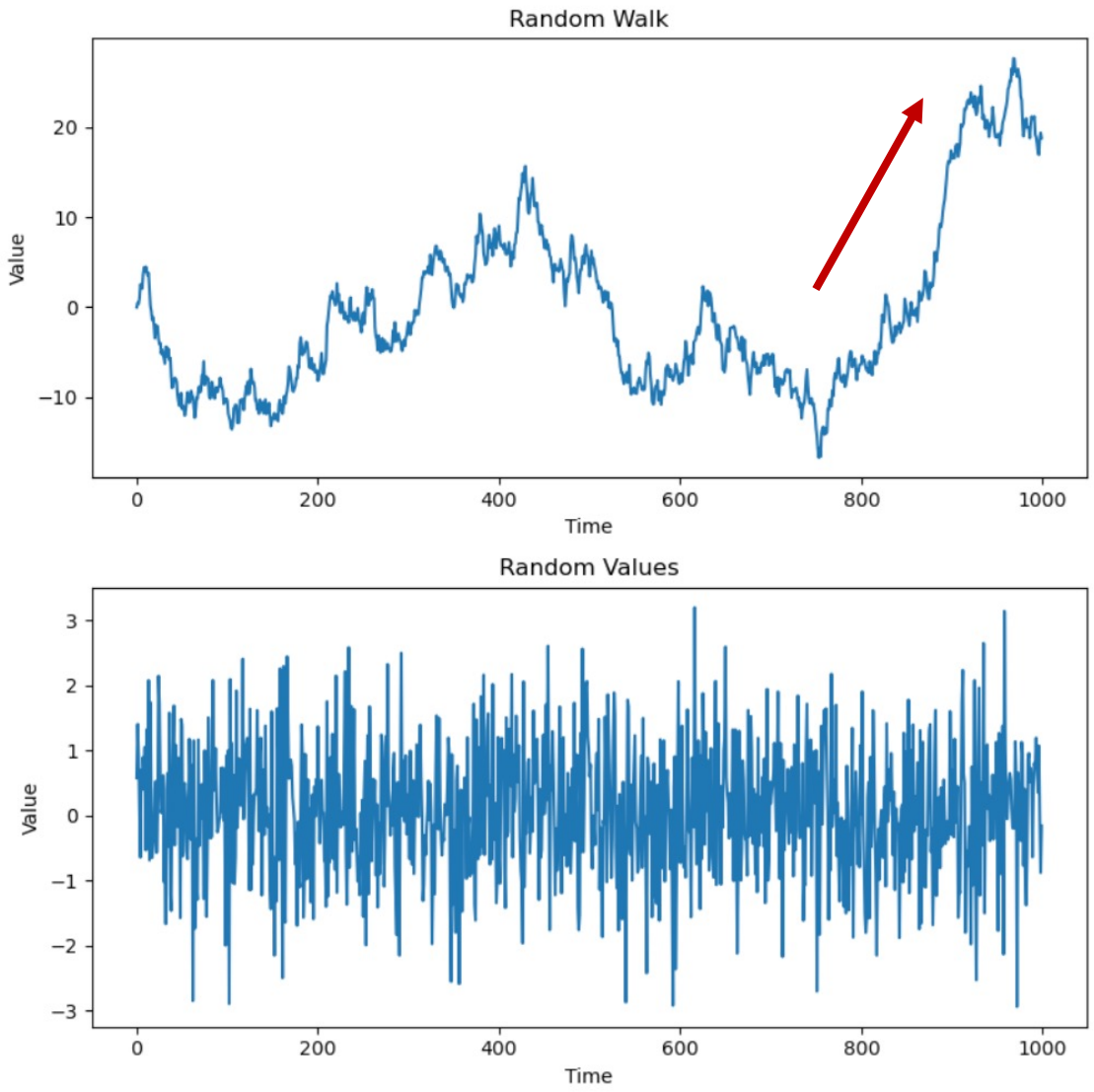

```
# Generate random walk
n_steps = 1000
walk = np.zeros(n_steps)
for i in range(1, n_steps):
    walk[i] = walk[i-1] + np.random.randn()
```



In random walk, each transition process is completely random, but the status of the next step (i) is affected by the previous step (i-1).

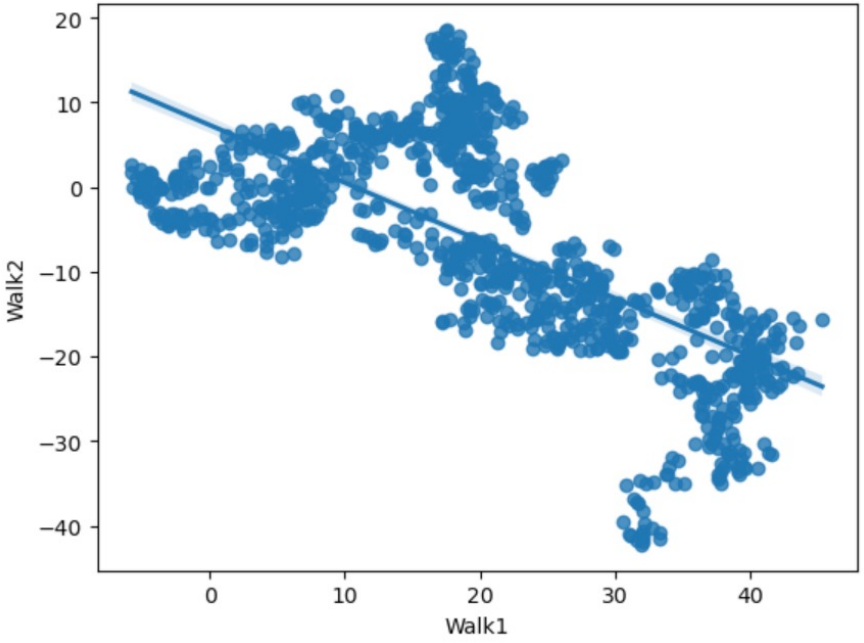
Interpretation of the trend

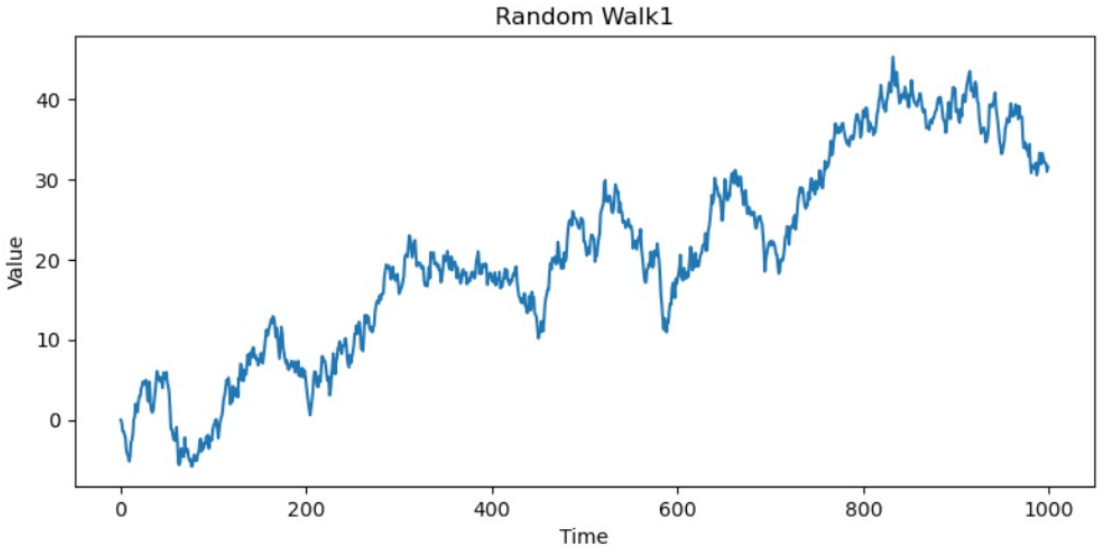
- If you find a trend in time series, do you think there is a specific mechanism to generate it?
- Not always. Random walk can generate the trend without a specific mechanism.
- This is very different from random values.

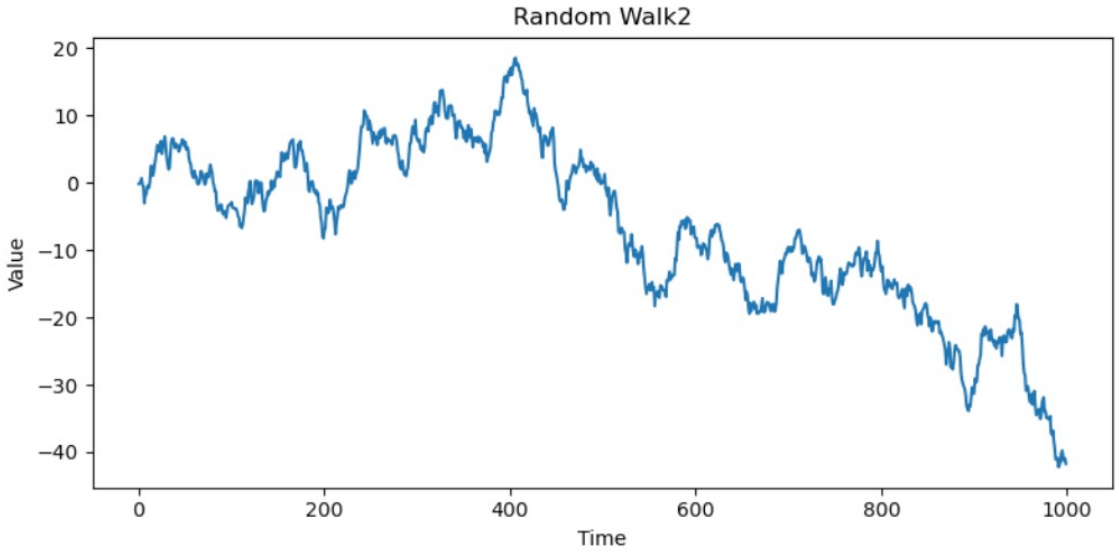


Interpretation of the trend

- Even if you find a significant "correlation" between two variables, it doesn't necessarily mean "causation".



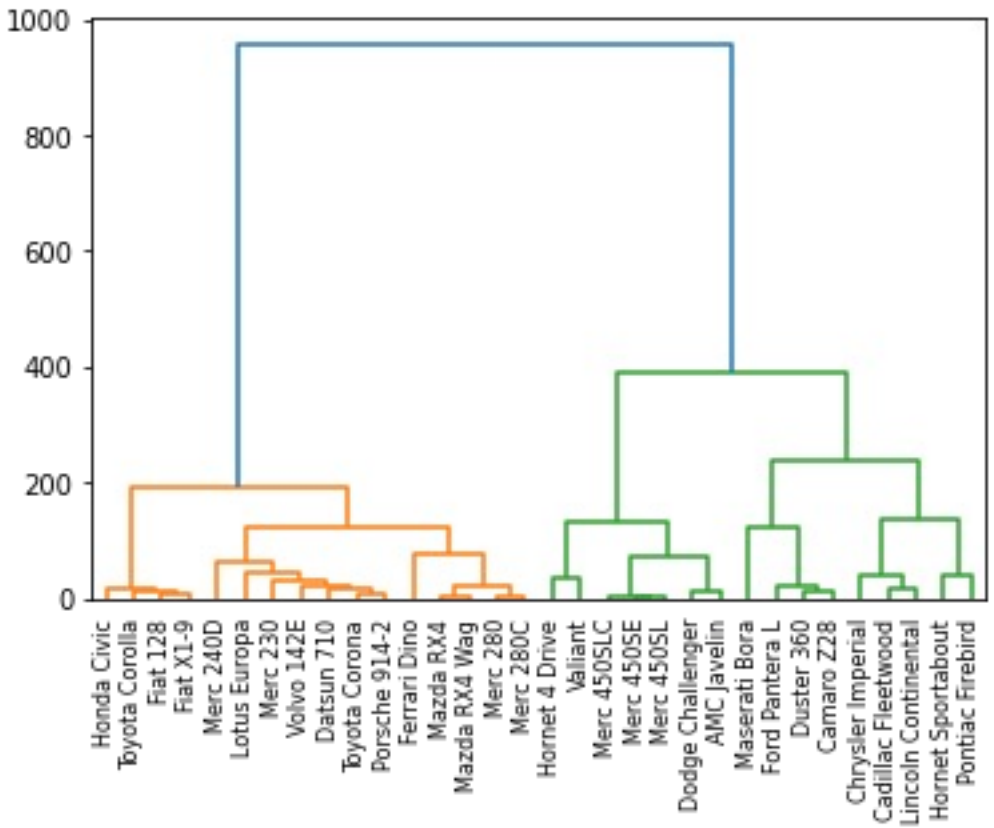




Dendrogram (Hierarchical clustering)

Dendrogram

- A **dendrogram** is a network structure. It is constituted of a root node that gives birth to several nodes connected by edges or branches.
- Dendrogram can be made with two types of dataset.
  - A numeric matrix allowing to compute **distance** between individuals
  - A **hierarchical** dataset where the relationship between entities is provided directly.

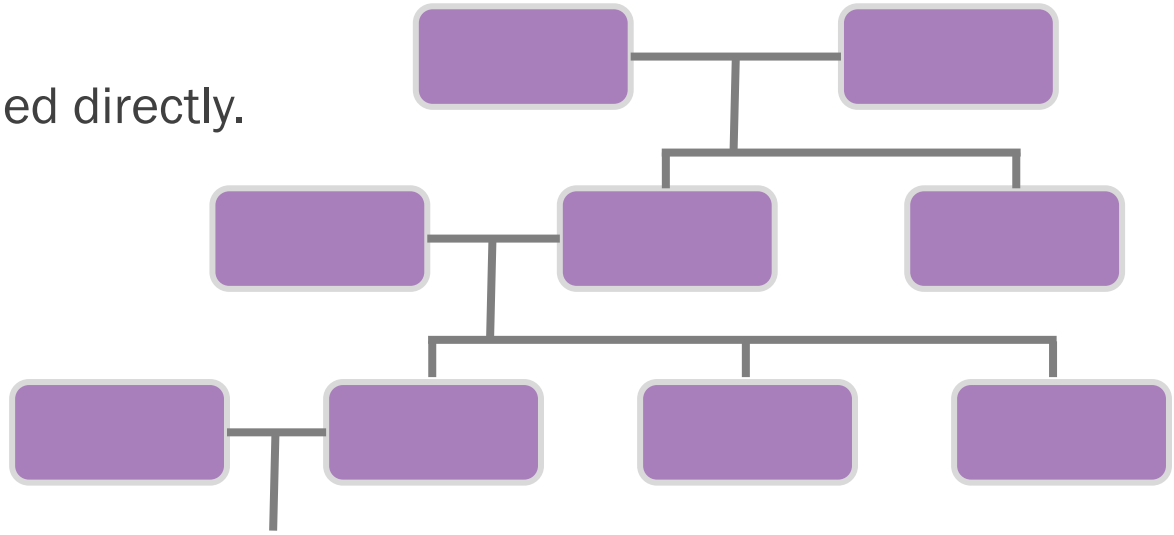


https://www.data-to-viz.com/#dendrogram

Dendrogram

Dendrogram can be generated from different types of data and methods; it is important to understand how it is generated.

- **Distance matrix**
  - Distance among data can be calculated from data (e.g., numeric data or sequence data). We can use the distance matrix to draw dendrogram (will be explained later).
- Hierarchical dataset
  - Sometimes the hierarchical structure is provided directly.
  - For example, a family tree.



https://www.data-to-viz.com/#dendrogram

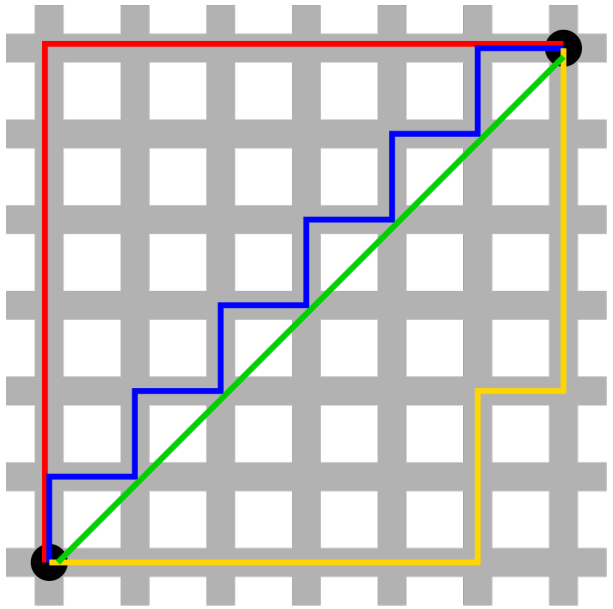
Hierarchical clustering

StatQuest (https://statquest.org/)

https://youtu.be/7xHsRkOdVwo?si=645miBM3aiQDQIKl

Dendrogram | How to calculate a distance matrix

- **Distance** is a measure of how far apart two points are (the amount of space between two points).
- There are many distance measures:
  - **Euclidean distance** (green line). Probably most commonly used. $d(p,q) = \sqrt{(p^2 - q^2)}$
  - **Manhattan distance** (red, blue, or yellow line). E.g., travel distance
  - **Edit distance**. A way to measure how dissimilar two strings (words) are.
    - Edit distance between "get" and "bat" is 2.
    - get ⟶ **b**et (replace "g" with "b")
    - bet ⟶ b**a**t (replace "e" with "a")
- There are other "non-distance" measures.



https://en.wiktionary.org/wiki/Manhattan_distance#/media/File:Manhattan_distance.svg

Dendrogram | How to calculate a distance matrix

- **Dissimilarity**
  - **Bray-Curtis dissimilarity**. Common and useful when measure compositional dissimilarity between two data (e.g., two ecological communities).

Site \ Species	Sp 1	Sp 2	Sp 3
Site A	80	90	120
Site B	150	100	50
Site C	30	25	20

Dissimilarity between Site A and B

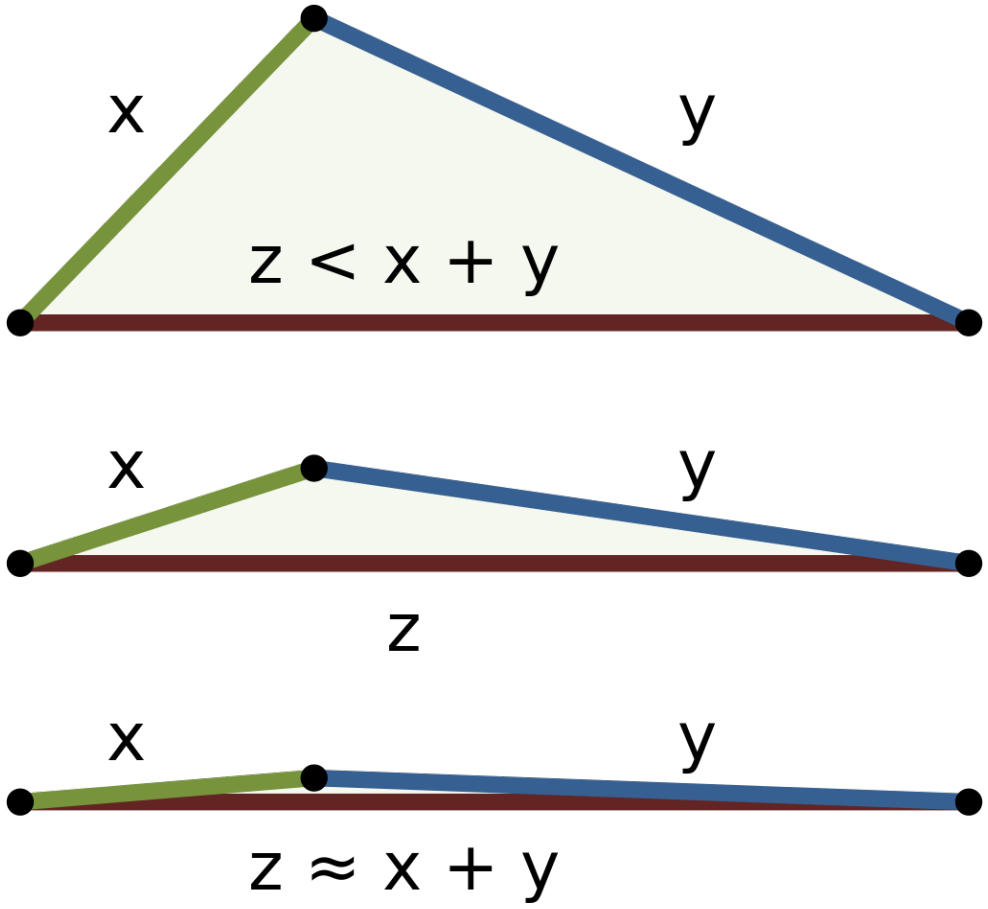
$$B_{AB} = \frac{\sum_{i=1}^{n} |A_i - A_i|}{\sum_{i=1}^{n} (A_i + B_i)}$$

$$B_{AB} = \frac{|A_1 - B_1| + |A_2 - B_2| + |A_3 - B_3|}{(A_1 + B_1) + (A_2 + B_2) + (A_3 + B_3)} = \frac{|80-150| + |90-100| + |120-50|}{(80+150)+(90+100)+(120+50)} = \frac{70+10+70}{230+190+170} = \frac{150}{590} \approx 0.25$$

If A and B are the same, B_{AB} = 0. If A and B are completely different, B_{AB} = 1. (=> Dissimilarity)
If you calculate 1 – B, it is “similarity index”.

Dendrogram | How to calculate a distance matrix

- **Distance and Dissimilarity**
  - The Bray–Curtis dissimilarity is often erroneously called a distance.
  - Bray-Curtis dissimilarity is **NOT** a distance measure.
  - It does not satisfy triangle inequality. It should always be called a dissimilarity to avoid confusion.
  - Mathematical definition:
    - *The distance between an object and itself is always zero.*
    - *The distance between distinct objects is always positive.*
    - *Distance is symmetric: the distance from x to y is always the same as the distance from y to x.*
    - *Distance satisfies the triangle inequality.*



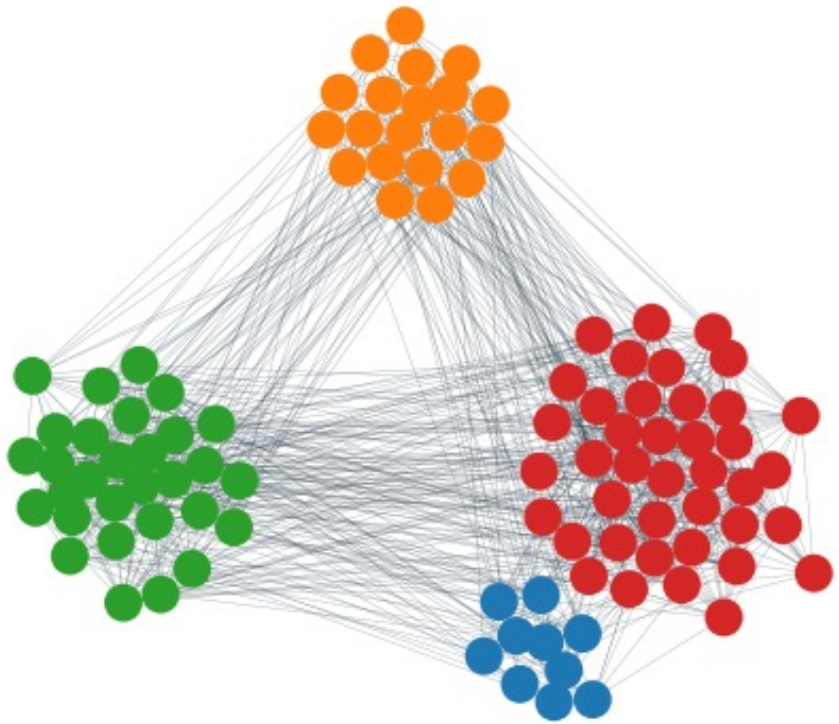
WhiteTimberwolf (CC-BY3.0)

https://en.wikipedia.org/wiki/Triangle_inequality#/media/File:TriangleInequality.svg

Network

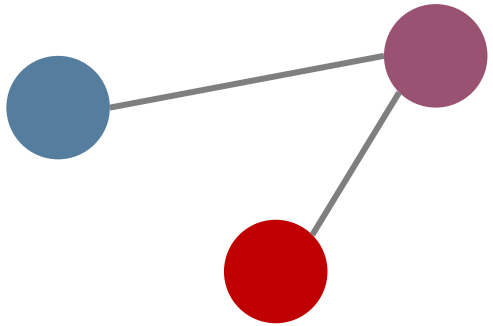
Network

- A **network** shows interconnections between a set of entities. Each entity is represented by a **node**. Connection between nodes are represented through **links** (or edges).
- "Network" is an entire field of research in itself. For example, you need to set up a layout algorithm that finds out an optimal position for each node.
- Several type of networks exist. They can be **directed** or **undirected**. Links can be **weighted** or not.

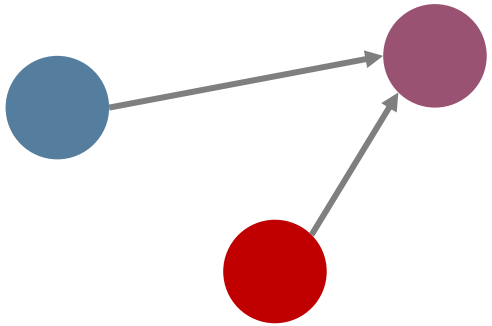


https://python-graph-gallery.com/500-network-chart-with-edge-bundling/

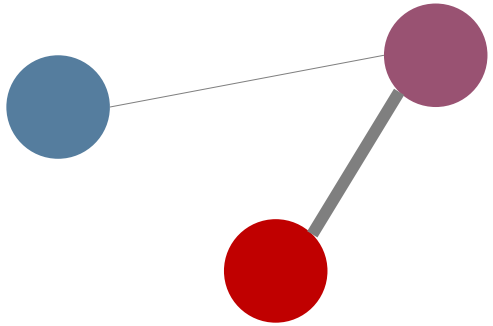
Network



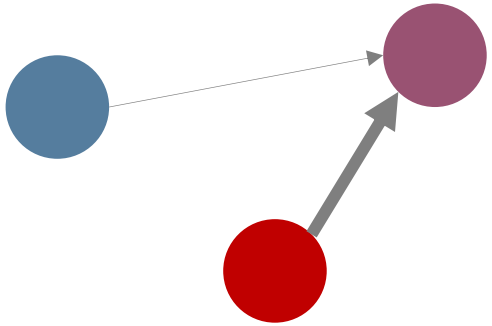
Unweighted, undirected network



Unweighted, directed network

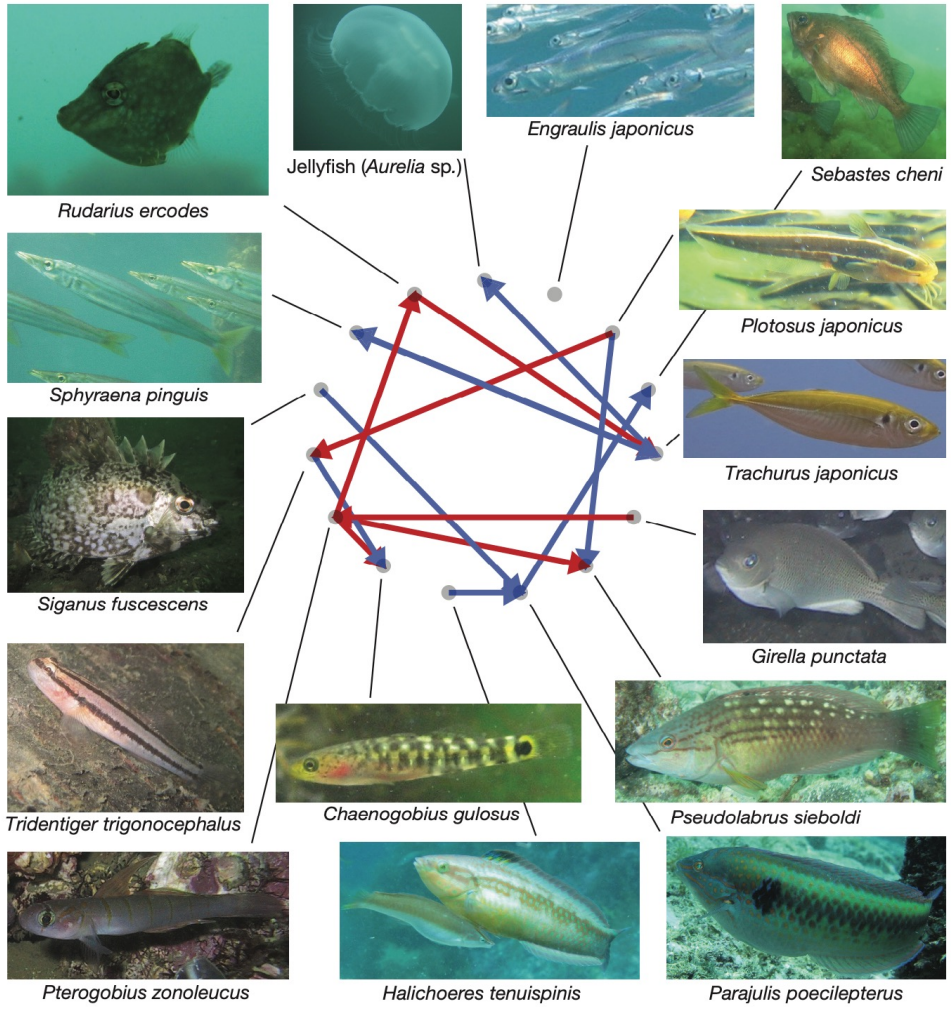


Weighted, undirected network



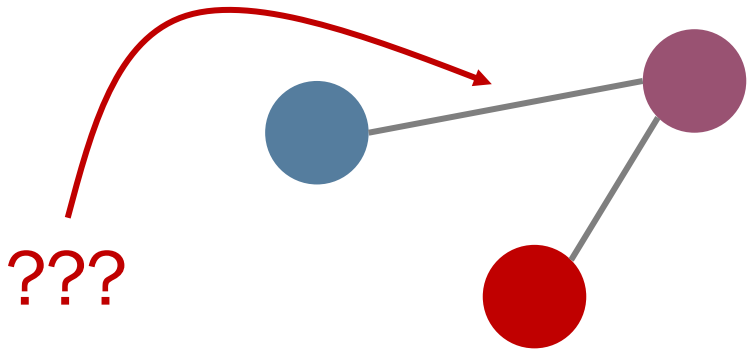
Weighted, directed network

Fish-fish interaction network



Ushio et al. (2018)

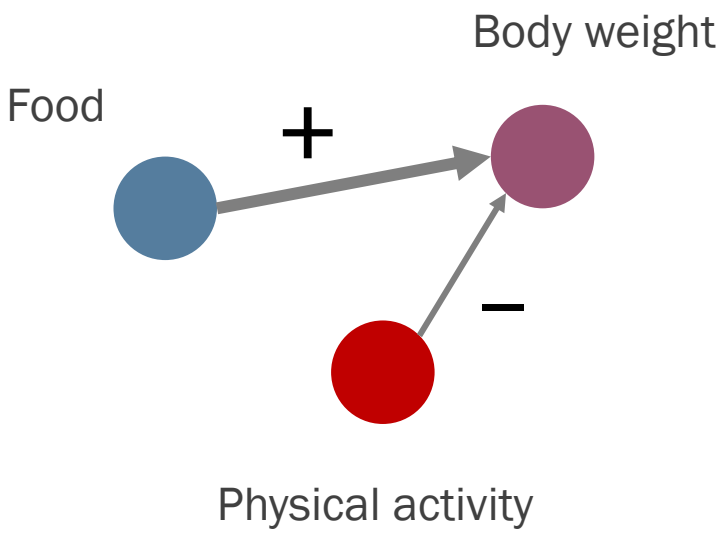
Network | How to define a linkage and weight

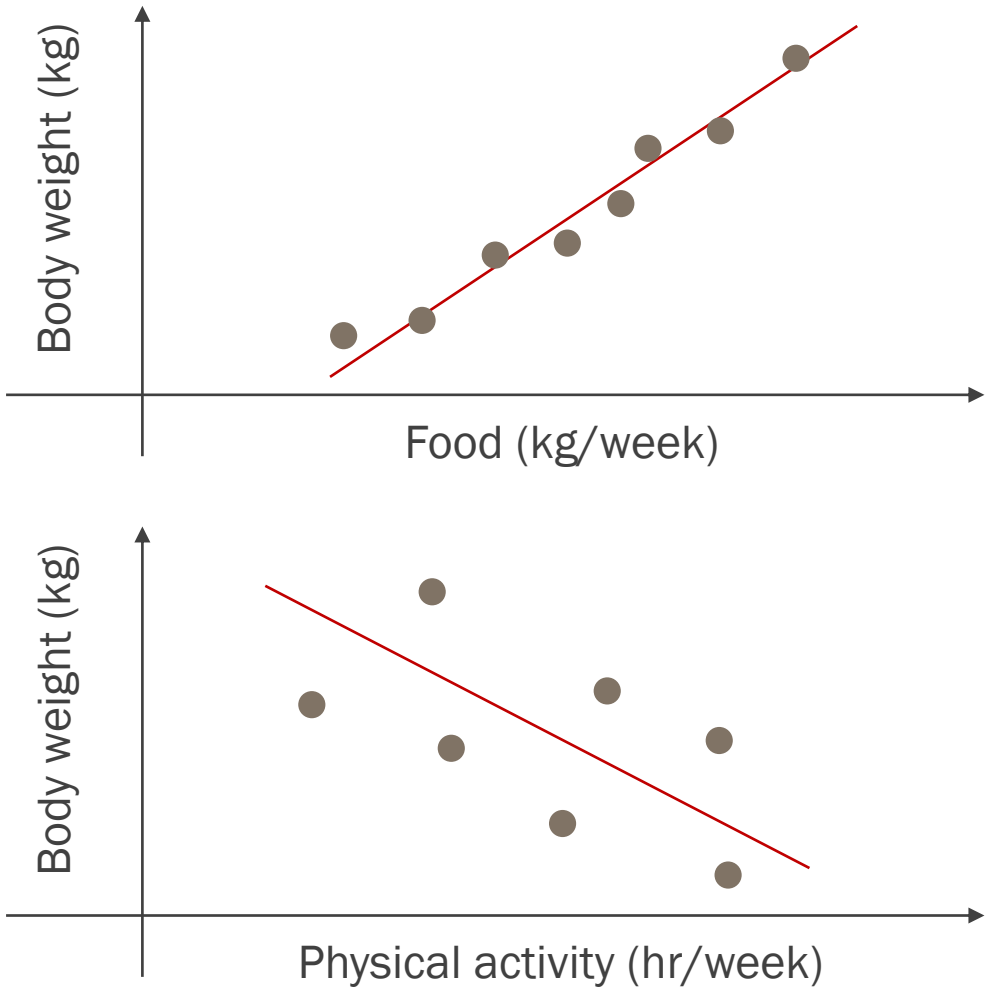


Unweighted, undirected network

- The concept of **network** is relatively simple, but how can we detect/define a linkage from data?
- There are also many ways:
  - Measure correlations (if the data is numeric)
  - Linkages are already known (e.g., SNS)
  - Mutual information, transfer entropy, … (too much!!)

Network | How to define a linkage and weight



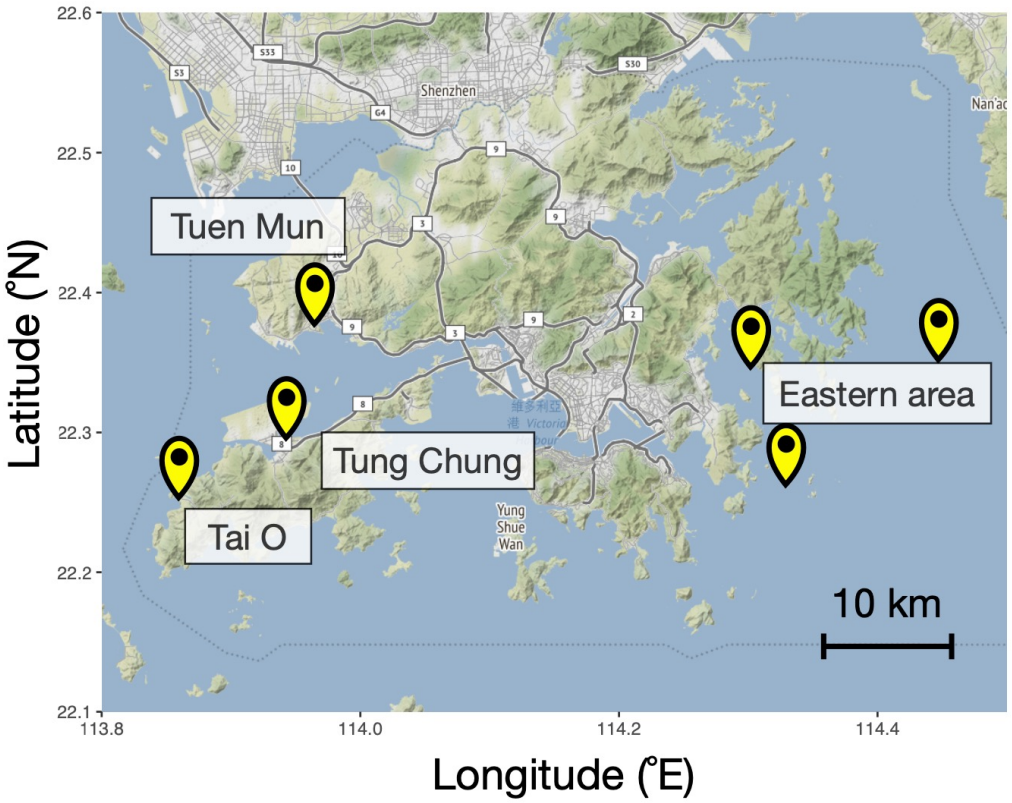


Map

Map

- **Map** is useful when you analyze geological, economics, political, and ecological data.
- We normally **do not** collect the data required to draw maps. Instead, we use (download) pre-existing data to draw maps.
- Thus, we need to care “terms and conditions” when we draw maps.

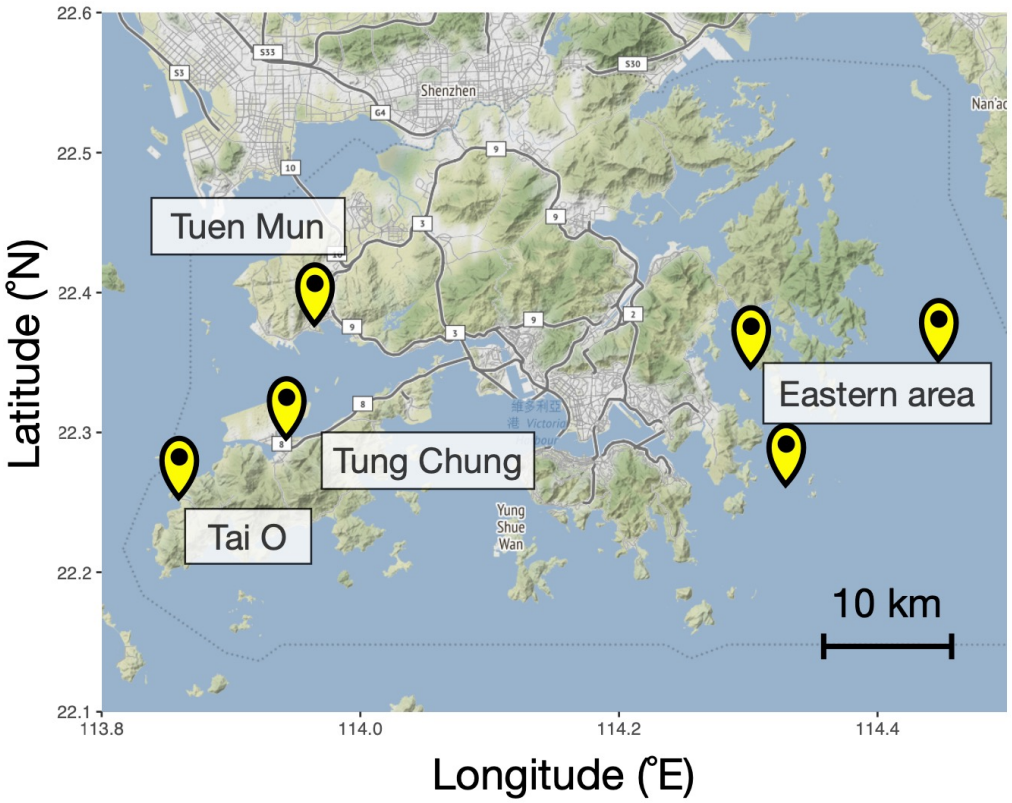
Drawn by “ggmap” package of R



Map | raster data and vector data

- **Raster** is made up of a grid of **pixels**, where each pixel represents a value.
- **Vector** is made up of individual **points**, **lines**, and **polygons**.
- Raster data is often used to create background maps that show the terrain, vegetation, or other continuous features.
- Vector data is often used to overlay points, lines, and polygons that represent specific features of interest, such as cities and roads.

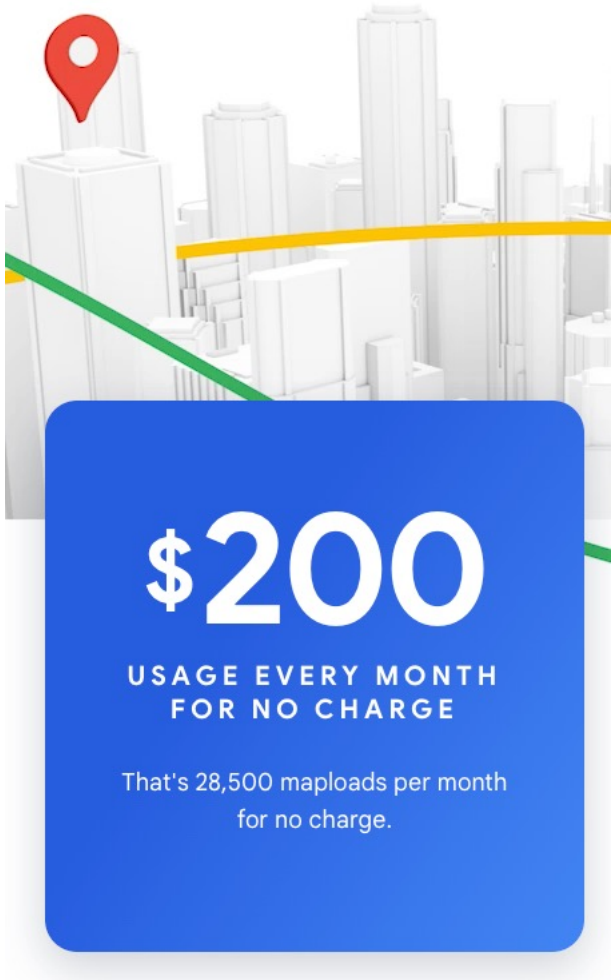
Drawn by “ggmap” package of R



Map resources

Google Maps

- The most famous one?
- You can embed the map in your product. (Strictly speaking, screenshot is maybe inappropriate…)
- But NOT completely open (free), but the terms and conditions are complicated…



For commercial use, see https://mapsplatform.google.com/pricing/

https://www.google.com/maps

Map resources

STADIA MAPS

https://stadiamaps.com/

Free	Starter	Standard	Professional
$ 0 /month	$ 20 /month	$ 80 /month	$ 250 /month
Hard limits enforced	3 ¢/thousand additional credits	2 ¢/thousand additional credits	1.5 ¢/thousand additional credits
200,000 credits / month	1,000,000 credits / month	7,500,000 credits / month	25,000,000 credits / month
Standard basemaps	Standard basemaps	Standard and satellite basemaps	Standard and satellite basemaps
Only routing and TZ APIs	Basic geospatial APIs	Advanced geospatial APIs	Professional geospatial APIs
Commercial use† *not* allowed	Commercial use† allowed	Commercial use† allowed	Commercial use† allowed
Get started today!	Get started today!	Get started today!	Get started today!

OpenStreetMap

https://www.openstreetmap.org/

© Open Data

OpenStreetMap is *open data*: you are free to use it for any purpose as long as you credit OpenStreetMap and its contributors. If you alter or build upon the data in certain ways, you may distribute the result only under the same licence. See the Copyright and License page for details.

License: ODbL 1.0

Map resources

OpenStreetMap

https://www.openstreetmap.org/

Copyright and License

OpenStreetMap® is *open data*, licensed under the Open Data Commons Open Database License (ODbL) by the OpenStreetMap Foundation (OSMF).

You are free to copy, distribute, transmit and adapt our data, as long as you credit OpenStreetMap and its contributors. If you alter or build upon our data, you may distribute the result only under the same licence. The full legal code explains your rights and responsibilities.

Our documentation is licensed under the Creative Commons Attribution-ShareAlike 2.0 license (CC BY-SA 2.0).

https://www.openstreetmap.org/copyright